

MAE 243: Final Project Technical Report

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Abstract

This study examines how cost uncertainties in emerging clean energy technologies shape optimal capacity expansion for a fully decarbonized WECC electricity system in 2045. A deterministic linear programming model evaluates fourteen scenarios combining cost reductions for enhanced geothermal systems, small modular reactors, and battery energy storage systems (BESS) under reference and high electrification demand. System costs decline across all cases as technologies become cheaper, with BESS cost reductions producing the largest changes in both cost and deployment. Geothermal expands only when costs fall by more than 50%, while nuclear capacity remains largely unchanged. High electrification scenarios show higher total costs and substantially greater storage deployment, emphasizing the role of flexible resources in meeting clean energy demand. Overall, results highlight the strong influence of technology learning trajectories on system design and investment priorities for a reliable, carbon-free WECC grid.

1 Introduction

1.1 Context and Motivation

The decarbonization of the electricity sector is widely recognized as a vital transformation necessary to meet national and international climate commitments in the coming years. In the United States, the Western Electricity Coordinating Council (WECC), which includes California, the Pacific Northwest, the Desert Southwest, and some Rocky Mountain States, serves as a critical region to ensure this transformation occurs. The regions within the WECC are also ably equipped to meet the challenge with many opportunities for low cost carbon free energy sources to meet electrical grid demand. Solar and Wind energies in particular serve as critical technologies in the decarbonization efforts. Outside of simply having the ability to decarbonize, many states have also enacted their own laws enforcing carbon free electric supplies in the next few decades, with the California mandate having a deadline of 2045.

A 100% clean electric grid in a large region like the WECC is a massive challenge. The region is home to a variety of climates and electrical needs, requiring a well designed energy system if demand is to be met safely, reliably, and

at a low cost. The minimization of costs, continuity of reliability, and elimination of emissions presents a variety of difficult constraints, one made even more difficult by the nature of the cheapest carbon free power sources. Variable Renewable Energy sources like Solar Photovoltaic (PV) and onshore wind have achieved cost reductions making them cheaper new energy sources than even conventional fossil fuel sources. However, innate to their name, their power production is variable, with outputs constrained by weather and diurnal cycles, creating periods of energy surplus and deficit. A fully carbon free grid that remains reliable will require complementary resources capable of providing firm, dispatchable capacity.

Three technologies are positioned as the primary candidates to fill the firm clean energy role in the WECC by 2045; nuclear power, enhanced geothermal systems (EGS), and battery energy storage systems. Nuclear power can provide high capacity baseload power with low variable cost, but incurs high capital costs for new capacity. The possibility of small modular reactors (SMRs) may reduce these capital costs, making nuclear more modular and reducing planning costs and construction delays. EGS is an emerging technology that could provide stable power from the Earth's internal heat, but its near term costs are still high, and commercialization relies on a fast pace of learning during initial large scale test projects. Battery energy storage, while not explicitly a "firm" energy source, has seen large cost reductions in the last decade and can serve a similar role to firm technologies. It is limited by the duration of activity, with most

modern installations cycling daily and limited by seasonal VRE.

The primary challenge motivating this study is that these three technologies are characterized by deep cost uncertainties extending into 2045. Present investment costs in the mid-2020s are defined by first of its kind project experiences, laboratory predictions, or early commercial demonstrations, which cannot be expected to reliably predict how mature system costs will look in the coming decades. The uncertainty of these costs could have major implications on each technologies role in a carbon free WECC, with large enough cost reductions having the potential to trigger large scale deployment. The works in this report seek to answer the question of how changing costs of clean energy technologies will affect the optimal deployment of carbon free power systems in a capacity expansion model. This report examines the basis of the cost reduction predictions of the three primary technologies, the capacity expansion models production and outputs, and a discussion of the results, their applicability, and future work.

1.2 Literature Review

Generation capacity expansion models have become the primary tool for evaluating long term investment decisions in electricity systems under decarbonization constraints. Key work comes in Sepulveda et al.[1], which established a result foundational to this work, that firm low carbon resources, will consistently reduce the cost of fully decarbonized electricity systems relative to those relying only on VRE and stor-

age. This result holds across a variety of geographic contexts and emission reduction goals, underscoring the value of dispatchable clean firm capacity.

Enhanced geothermal systems have historically been excluded from macro scale decarbonization studies due to the uncertainty of their costs and geographical constraints necessary for utility scale deployments. Recent test demonstrations, such as those by Fervo Energy in Nevada and Utah, and the U.S. Department of Energy’s FORGE project, have generated some of the first ground truth cost data for modern EGS development. Work by Ricks and Jenkins, [2] combine this field data with an experience curve framework to project EGS cost trajectories and deployment pathways through the year 2050 across the continental U.S. Their central finding is that EGS deployment is highly sensitive to the pace of cost reduction, with cost reductions of above 70% potentially enabling widespread dispatch. Ricks and Jenkins also determine that geothermal is well suited to western states where subsurface temperature gradients are particularly steep and that initial deployment will be easier in those regions. The experience curve projections from Ricks and Jenkins (2025) directly inform the EGS cost scenarios constructed in this study.

The cost of new nuclear capacity in the U.S. has been a hot button issue for years, with issues such as budgetary constraints, scheduling issues, waste containment, and environmental impact all keeping most large scale projects from being completed and forcing the shuttering of pre-existing installations. Nuclear has re-

cently gained traction in the national conversation over energy once more, and SMRs could reduce many of the complications associated with large nuclear installations if done properly. The NREL Annual Technology Baseline (ATB) from 2024 [3], provides the most comprehensive publicly available cost projections on new nuclear through the year 2050, and distinguishes large light water reactors and SMRs under conservative, moderate, and advanced technology innovation scenarios. The ATB applies historically observed learning rates to projected deployment scenarios and generates estimates of cost recutations for these technologies, yielding reductions from 30% in conservative to 61% in advanced for SMR nuclear. Nuclear capacity decisions are often limited by costs, but the level of uncertainty is high for how nuclear costs will develop in the coming years, and high reductions being completely possible.

Utility scale battery storage costs have declined more quickly than projections consistently in the last few years, a result of widespread lithium ion cell manufacturing and improvements in system setups reducing costs dramatically. The NREL ATB 2024 report also includes estimated cost reductions of battery energy storage systems (BESS) ranging from 20% in conservative scenarios to 69% in advanced cases. This technologies revolution is already in full swing thanks to well established manufacturing industries and markets for these devices. Generally, modern markets are dominated by short storage (4 hour) systems, but with solar PV and BESS becoming a more popular utility scale pairing, this timescale may also trend up-

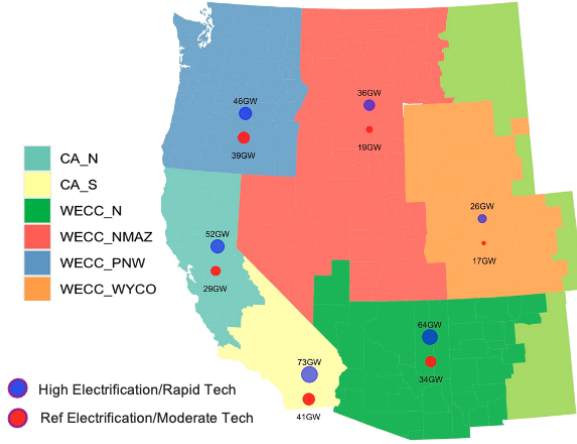


Figure 1: 6 zones within the WECC with relative demand for reference and high electrification cases shown via bubble size.

wards in the coming years.

2 Data

This model utilized a dataset provided by the instructor for the course. The dataset provides inputs for capacity expansion studies for a carbon free electricity system in the WECC U.S. regions. Public data was utilized to define necessary parameters within the dataset. Only resources that do not emit carbon dioxide during operations were included in the dataset and this model. Existing capacity for VREs, hydro related energy facilities, and nuclear is included for all states in the examined region. The dataset includes 16 representative weeks with hourly resolution for each week and utilizes weights to minimize error in each time series within the model.[4] The model contains 6 zones defined within the WECC as shown in Figure 1. 159 generators are defined in the dataset across the 6 zones. The dataset also includes a refer-

ence electrification case and a high electrification case for the model, both of which are based on a 2018 NREL study[5]. Both cases of electrification were ran through the model with differences examined in the Results and Discussion. As mentioned in the introduction, three technologies were chosen to model how cost reductions impact deployment, with each technology having costs for conservative, moderate, and advanced technological innovation scenarios. The percentages of cost reductions, applied to capital and fixed operation costs, are used to simulate the cost reductions found in Ricks and Jenkins and the NREL ATB 2024 report. The relative conservative, moderate, and advanced scenario cost reduction percentages are tabulated below in Table 1[2][3].

Geothermal multipliers are grounded within the projections in Ricks and Jenkins. The conservative scenario (multiplier 0.8) corresponds to their finding that reductions below 50% lead to minimal EGS deployment. Testing this logic the moderate scenario is exactly a 50% reduction. Lastly, the optimistic scenario tests the key finding of the paper with a reduction over 70%, to investigate how the calibrated threshold from Ricks and Jenkins for widespread EGS dispatch. The remaining values for nuclear and battery storage are pulled directly from the NREL ATB. Using varied combinations of the 3 unique scenarios for each technology, 14 scenarios were defined for testing, which are explored completely in the methods section.

This model utilizes the full dataset as provided sans the flexible load options. This additional data was intended to model how the shift-

Technology	Conservative	Moderate	Advanced
Geothermal (EGS)	20%	50%	72%
Nuclear (SMR)	30%	44%	61%
BESS	20%	52%	69%

Table 1: Cost Reduction Multipliers for the 3 selected technologies.

ing of electrical loads could improve results, but was not relevant for this model. The ways data was input to produce the results in explored fully in the Methods.

3 Methods

3.1 Model Overview

This project uses a deterministic linear programming (LP) capacity expansion model implemented in Julia using the JuMP framework with the Gurobi LP solver. The model takes the 2045 all clean energy scenario for the WECC region and seeks to find the cost minimizing portfolio best suited to generation investment, storage deployment, transmission reinforcement, and hourly dispatch. New build fossil fuel generators are banned and they are also phased out from existing resources. The planning horizon is a single year, 2045, and utilizes the datasets brownfield representation of the future western interconnection in the U.S. This model resolves three categories of decisions, investment planning, early retirement, and hourly dispatch.

3.2 Mathematical Formulation

The mathematical formulation for this project follows standard capacity expansion conventions. All variables, both those defined and not defined in the body of this report, are defined in the appendix. Decision variables for generators include new power capacity for generators, total installed power capacity (existing + new - retired), early retirement of existing generators, new storage capacity built, and early retirement of storage capacity. Transmission and dispatch decision variables are total transmission on lines in the model, new transmission added, power dispatch of generators at an hourly rate, storage charging and state of charge (SoC), net power flow on given lines, and the non-served energy (NSE) in specific segments of zones in a given hour. These decision variables and their units are tabulated in Table 2. This is fairly standard to capacity expansion models and does not do anything significantly unique compared to prior literature.

Moving on to key parameters, most of these come directly from the dataset, including annualized investment costs for generators and for storage, fixed operations and maintenance (O&M) costs for generators, storage, and exist-

Parameter / Variable	Definition	Units
x_g^{new}	New power capacity built	MW
x_g^{cap}	Total installed power capacity (existing – retired + new)	MW
e_g^{new}	New storage energy capacity built	MWh
e_g^{cap}	Total storage energy capacity	MWh
T_l^{cap}	Total transmission capacity on line l	MW
T_l^{new}	New transmission capacity added to line l	MW
$P_{g,t}$	Power dispatch of generator g at hour t	MW
$u_{s,z,t}$	Non-served energy, segment s , zone z , hour t	MW
C_g^{inv}	Annualized investment cost	\$/MWyr
C_g^{fom}	Fixed operations & maintenance cost	\$/MWyr
$C_g^{\text{inv,e}}$	Annualized storage energy investment cost	\$/MWyr
$C_g^{\text{fom,e}}$	Fixed O&M on storage energy capacity	\$/MWyr
C_g^{var}	Variable O&M + fuel cost per MWh generated	\$/MWh
C_l^{fix}	Fixed cost of existing transmission capacity	\$/MWyr
C_l^{inv}	Investment cost of new transmission capacity	\$/MWyr
w_t	Sample weight (annual hours represented by modeled hour t)	h/yr
V_s	Value of lost load for NSE segment s	\$/MW

Table 2: Variable Definitions for the Objective Function

ing transmission capacity, investment costs for new transmission, hourly capacity factors for variable renewable energy (VRE) generators, electrical demand in a given zone on an hourly basis, storage charge and discharge efficiency, ramp rates for generators, brownfield installed capacity, resource ceilings for new builds, the value of lost loads for NSE, maximums for demand curtailment in specific segments of the model, and lastly the network interconnection matrix. These are also tabulated in Table 2.

Looking now at the objective function, the objective function is the minimization of the function Z , the total annualized system cost. Z is the sum of 4 cost components including the fixed investment and operations and mainte-

nance(O&M) costs, weighted operational costs, non-served energy costs, and transmission costs. (Equations 1-5)

$$\min Z = Z_{\text{fixed}} + Z_{\text{var}} + Z_{\text{NSE}} + Z_{\text{transmission}} \quad (1)$$

$$Z_{\text{fixed}} = \sum_g \left(C_g^{\text{inv}} x_g^{\text{new}} + C_g^{\text{fom}} x_g^{\text{cap}} + C_g^{\text{inv,e}} e_g^{\text{new}} + C_g^{\text{fom,e}} e_g^{\text{cap}} \right) \quad (2)$$

$$Z_{\text{var}} = \sum_t \sum_g (w_t C_g^{\text{var}} P_{g,t}) \quad (3)$$

$$Z_{\text{NSE}} = \sum_t \sum_s \sum_z (w_t V_s u_{s,z,t}) \quad (4)$$

$$Z_{\text{transmission}} = \sum_l (C_l^{\text{fix}} T_l^{\text{cap}} + C_l^{\text{inv}} T_l^{\text{new}}) \quad (5)$$

The fixed generation cost component includes investment and fixed O&M for both power capacity and storage energy capacity. Variable operational costs are weighted by the sample weight w_t of each modelled hour, which properly scales the representative period costs to annual totals. For all VRE, storage, geothermal, hydro, and nuclear resources, variable costs are either zero or near zero as defined by the dataset. The non-served energy component penalizes the value of lost load for each demand segment while solving. The transmission costs cover both fixed costs for existing lines and any investment costs necessary for reinforcement of the grid.

To ensure that the model does not drift too far from reality while running, constraints were put in place to guide the system toward more realistic results. Constraints served purposes like ensuring the total installed capacity was equal to the existing capacity at the start, and then followed and accounted for retirements and new builds properly. Energy balances were also important to ensure demand was met, that generation was not above the total capacity, and that ramp rate limits were not exceeded between time windows. This occurs in every zone at every modeled hour, with interzonal flows represented via the network incidence matrix $A_{l,z}$. The energy balance dual variable is $\lambda_{z,t}$, which produces the locational marginal price in each zone and hour, quantifying the marginal cost of serving incremental load. Generation limits and must run constraints were also put in place for generators to ensure generation does not exceed capacity (VRE sources) or below minimum sta-

ble generation rates. (Geothermal, hydro, and biomass) Along with this ramp rates.

Battery dynamics were also constrained to ensure the state of charge is contiguous between hours and that battery behaviors like only charging or discharging at a single time or that battery discharge capacity was not exceeded in the model. Transmission limits were placed to ensure power flows are bounded by transmission capacity. Lastly, curtailment is constrained to cap it at a fraction of the local demand based on a willingness to pay for the costs within the model.

3.3 Cost Scenario Design

The key addition to this project that puts it above merely modeling the data as provided is the cost scenario modeling. The goal of this project is to investigate how sensitive the model results are to cost reductions with 3 key technologies, geothermal, specifically enhanced geothermal systems (EGS), nuclear, with a focus on possible cost reductions small modular reactors may induce on the industry, and battery energy storage systems. These 3 generation/ storage methods are all currently fairly expensive and can be hard to justify, yet all 3 have a growing movement behind them that may lead to cost reductions above projections. Geothermal and nuclear have the ability to serve as clean baseline loads for all clean energy grids that could massively reduce the need to overbuild on VREs like solar and wind. If these technologies become cost competitive enough that could drive down overall prices for a capacity expansion model like

this one. Batteries, while serving a different purpose than geothermal and nuclear, can also aid in reducing the need to overbuild on VRE, and reduce curtailment during high renewable periods.

The 14 scenarios produced by this model systematically vary the investment and fixed O&M costs for these technologies by applying scalar multipliers to the baseline values defined in the dataset. The dataset's baseline costs reflect a 2045 WECC prior to any scenario adjustments and represent a plausible estimate of technology costs under moderate learning trajectories. In this project, multipliers below 1.0 represent cost reductions relative to the baseline; for example a multiplier of 0.5 is a 50% reduction of the technologies investment and fixed O&M costs, and a multiplier of 0.8 is a 20% reduction.

The scenarios generally follow 2 methods. The first examines conservative, moderate, and advanced scenarios for each technology individually, with a multiplier applied to only 1 technology at a time, representing 9 scenarios. The second variety examines how these reductions work together, with a conservative, moderate, and advanced scenario where all 3 technologies have their respective multipliers for each scenario applied. In addition to this, 2 cross scenarios pair a more varied mix, with one utilizing geothermal optimistic and nuclear pessimistic, and the other being nuclear and BESS optimistic. This is to further explore how complementary dynamics may pair together to influence the results. The 14 scenarios are tabulated explicitly in Table 3.

3.4 Model Assumptions

This model relies on several simplifying assumptions that shape how the optimization behaves and how the results should be interpreted. First, the model operates under perfect foresight, hourly demand, VRE availability, and all operational conditions are assumed to be known in advance for the entire modeled year. The model is deterministic and does not incorporate uncertainty in weather, outages, or technology performance. Second, the temporal structure uses sixteen representative weeks to approximate annual variability, which captures broad seasonal patterns but cannot fully represent extreme events or long-duration reliability challenges. Third, generator performance is simplified to fixed capacity factors for VRE and fixed O&M and investment costs for all technologies, with no iterative learning or degradation effects. Fourth, the transmission network is represented using six aggregated WECC zones with interzonal transfer limits, meaning that interzonal congestion, siting constraints, and localized deliverability issues are not modeled. Fifth, the optimization is cost only and does not include reliability metrics, which may undervalue firm resources relative to VRE and storage. Finally, the model assumes that new generation and storage can be built up to the resource ceilings provided in the dataset without land use, permitting, or siting constraints. These assumptions allow the model to focus on the economic effects of cost reductions but also limit the direct policy applicability of the results.

Scenario	Multiplier(s) [EGS mult., Nuclear mult., BESS mult.]
EGS Conservative	[0.5,1,1]
EGS Moderate	[0.5,1,1]
EGS Advanced	[0.28,1,1]
Nuclear Conservative	[1,0.7,1]
Nuclear Moderate	[1,0.56,1]
Nuclear Advanced	[1,0.39,1]
BESS Conservative	[1,1,0.8]
BESS Moderate	[1,1,0.48]
BESS Advanced	[1,1,0.31]
ALL Conservative	[0.8,0.7,0.8]
ALL Moderate	[0.5,56,0.48]
ALL Advanced	[0.28,0.39,0.31]
EGS Advanced, Nuclear Conservative	[0.28,0.7,1]
Nuclear Advanced, BESS Advanced	[1,0.39,0.31]

Table 3: Scenarios with cost reduction multipliers tabulated in a matrix.

3.5 Implementation

The model is implemented via Julia using JuMP for the algebraic model construction and Gurobi as the LP solver. The full model comprises approximately 573,000 decision variables and 1.6 million constraints per scenario, dominated primarily by generation limit constraints and ramp rate constraints. Gurobi’s interior point algorithm is used for all model solves, which is useful since the model couples time with storage and ramp constraints. A 2 hour time limit was set for each 14 scenario run for the reference and high electrification datasets. This constraint was never enforced as the 14 scenario runs took roughly 70-85 minutes each. Results from each scenario are written to file in a catalog folder. A

pilot run that reduces to 2 of the 6 zones, fewer generators, and a smaller temporal window, is also included in the project code file.

With the model structure, cost scenario design, and dataset fully defined, the framework is now in place to evaluate how the system responds to changes in technology costs under both reference and high electrification scenarios. The 14 scenarios described above allow the model to explore a wide range of plausible cost trajectories for geothermal, nuclear, and battery storage, and to reveal how these trajectories influence investment decisions, dispatch behavior, and overall system cost.

Scenario	Total Cost (\$B/yr)	Geo Cap (GW)	Nuc Cap (GW)	BESS Cap (GW)
EGS Conservative	13.895/ 20.087	0.643/ 0.659	5.166/ 5.166	2.559/ 23.276
EGS Moderate	13.839/ 20.030	1.650/ 1.650	5.166/ 5.166	2.559/ 23.276
EGS Advanced	13.790/ 19.981	1.650/ 1.650	5.166/ 5.166	2.559/ 23.276
Nuclear Conservative	13.739/ 19.931	0.000/ 0.016	5.166/ 5.166	2.559/ 23.276
Nuclear Moderate	13.666/ 19.858	0.000/ 0.016	5.166/ 5.166	2.559/ 23.276
Nuclear Advanced	13.578/ 19.770	0.000/ 0.016	5.166/ 5.166	2.559/ 23.276
BESS Conservative	13.814/ 19.682	0.000/ 0.000	5.166/ 5.166	6.672/ 26.598
BESS Moderate	13.337/ 18.701	0.000/ 0.000	5.166/ 5.166	15.419/ 41.576
BESS Advanced	12.973/ 17.885	0.000/ 0.000	5.166/ 5.166	19.124/ 74.045
ALL Conservative	13.657/ 19.525	0.643/ 0.659	5.166/ 5.166	6.672/ 26.629
ALL Moderate	13.054/ 18.415	1.650/ 1.650	5.166/ 5.166	15.408/ 41.566
ALL Advanced	12.552/ 17.462	1.650/ 1.650	5.166/ 5.166	19.124/ 74.047
EGS Adv + Nuclear Cons	13.634/ 19.824	1.650/ 1.650	5.166/ 5.166	2.559/ 23.276
Nuclear Adv + BESS Adv	12.655/ 17.567	0.000/ 0.000	5.166/ 5.166	19.124/ 74.045

Table 4: System Cost and Installed Capacity by Scenario. (Reference data/ High Electrification Data)

4 Results

The 14 scenarios ran for each of the 2 datasets produced generally similar trends with slight variations. The total cost of each scenario, with results listed as (reference dataset result) / (high electrification dataset result), and summed capacity across the interconnection for the 3 primary technologies are shown in Table 4. The geothermal, nuclear, and BESS generation totals across the 6 zones for the year are compiled in Table 5. Across all scenarios, as technological advancement increases, thus decreasing investment and fixed O&M costs for the selected technologies, the total cost of the system decreases. This is a fairly straightforward result and does not suggest much as to how the scenarios relative reductions of cost actually affected the opti-

mal result.

High electrification dataset scenarios see a higher cost on the order of \$6 B/year relative to the reference case. The technology most responsive to the decrease of costs was BESS. Scenarios with BESS cost reductions saw the largest difference in cost and in installed capacity for the technology within the WECC, across both reference and high electrification scenarios.

Installed capacity results show distinct shifts across technology groupings. Geothermal increases to 1.65 GW only in scenarios where EGS costs are modified at or above 50%. In scenarios without an EGS cost modification, the nuclear and BESS scenarios, not only is there no new geothermal capacity, but the model also retires existing geothermal capacity early to com-

Scenario	Geo (TWh/yr)	Gen	Nuc (TWh/yr)	Gen	BESS Dis-charge (TWh/yr)
EGS Conservative	5.27/ 5.15		0.06/ 0.08		4.11/ 26.97
EGS Moderate	13.31/ 13.06		0.06/ 0.08		8.56/ 20.20
EGS Advanced	12.61/ 13.02		0.06/ 0.08		4.41/ 26.77
Nuclear Conservative	0.00/ 0.09		0.06/ 0.08		12.81/ 25.69
Nuclear Moderate	0.00/ 0.08		0.06/ 0.08		4.28/ 32.33
Nuclear Advanced	0.00/ 0.06		0.06/ 0.08		5.21/ 75.48
BESS Conservative	0.00/ 0.00		0.07/ 0.06		6.92/ 14.56
BESS Moderate	0.00/ 0.00		0.11/ 0.08		12.19/ 28.95
BESS Advanced	0.00/ 0.00		0.12/ 0.16		23.56/ 40.58
ALL Conservative	5.15/ 4.68		0.07/ 0.06		9.14/ 15.21
ALL Moderate	12.86/ 12.36		0.11/ 0.08		14.08/ 31.23
ALL Advanced	12.75/ 12.87		0.12/ 0.16		22.57/ 44.94
EGS Adv + Nuclear Cons	13.27/ 13.35		0.06/ 0.08		3.28/ 27.72
Nuclear Adv + BESS Adv	0.00/ 0.00		0.12/ 0.16		21.63/ 35.02

Table 5: Installed Annual Generation Summary by Scenario. (Reference data/ High Electrification Data)

pletely eliminate the technology from the mix. Nuclear capacity remains fairly unchanged regardless of the cost modifiers, with no scenario resulting in a unique Nuclear capacity value. BESS capacity varies widely, with modest deployments in reference EGS and Nuclear only scenarios at 2.559 GW. This changes substantially in high electrification trials, jumping up to 74.045 GW of installed capacity in scenarios with the highest cost reduction for the technology.

Generation results largely follow the installed capacity patterns. Geothermal sees generation generally increase as the cost goes down, but does not see a significant difference in results between the reference and high electrification dataset trials. Nuclear remains low in generation in all scenarios, with increasing generation

corresponding to a decrease in BESS costs, otherwise generation remains flat across trials. For BESS, high electrification scenarios saw BESS discharge increase sharply across all scenario categories, with the highest discharge over the scenario period occurring in advanced cases for BESS and nuclear.

Examining the rest of the system now, (See Figures 2 and 3) across datasets and scenarios solar PV was the dominant generator. This resource consistently made up the largest share of generation by a significant margin too. On-shore wind and conventional hydro are the other 2 biggest players fairly consistently. Generally biomass and gas + carbon capture and storage make up very little to no amount of the mix. For the reference data trials the total generation capacity stays fairly consistent across sce-

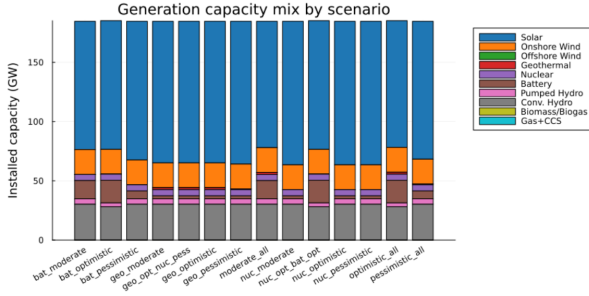


Figure 2: Reference Dataset scenarios Generation mix (Conservative = Pessimistic, Moderate = Moderate, Advanced = Optimistic)

narios, with some variation occurring in how the total is achieved, but the actual value varying only negligibly for any given scenario. When the mix varies, this is generally achieved by decreasing the total solar deployment, with other resources remaining relatively similar in capacity across scenarios. The high electrification datasets showcases a slightly different story, with a greater mix in total capacity occurring when BESS costs are reduced via the advanced trial multiplier. A slight increase above the average otherwise occurs in trials with moderate BESS cost reduction, but the change is most prominent in advanced scenarios. When these changes occur the capacity of any other given resource remains fairly stable, with total generation just increasing rather than any resource getting replaced when another grows in capacity.

Additionally, while not tabulated in the table or figures, this model also calculated NSE and transmission expansion for each trial. These results were generally insignificant as the complexity of the model allows for zero NSE across all trials, meaning the grid was able to perfectly meet demand in each case for all hours. Transmission also generally saw no new builds for

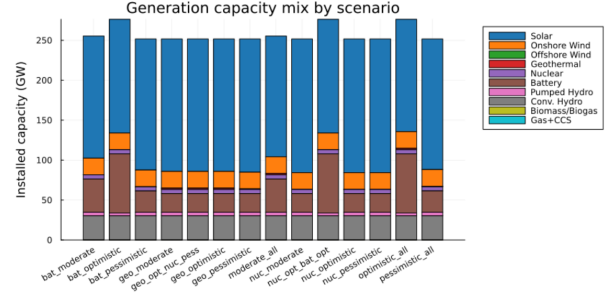


Figure 3: High Electrification Dataset scenarios Generation mix (Conservative = Pessimistic, Moderate = Moderate, Advanced = Optimistic)

transmission between the 6 zones. In the rare case that it did, new transmission build out costs were on the scale of 1-5% of the total cost and seen as low enough to limit reporting on these results.

These results have some key takeaways to be discussed further in the next section. First, as each given technology advances and the relative cost to deploy and operate a generator of its variety decreases, the total costs for the entire interconnection go down. In the reference data set trials, pairing all the possible cost reductions brings down the total costs from 13.895 \$B/year to 12.552 \$B/year. In high electrification trials the reduction is even more prominent, going from 20.087 \$B/year to 17.462 \$B/year. Secondly, despite these decreases in total system cost, the mix of generation type does not appear to change significantly as costs go down for geothermal and nuclear technologies, but the mix changes slightly as BESS costs go down. For this model, the reduction of costs for geothermal and nuclear is not enough to encourage any significant change in capacity mix expanding the presence of firm carbon free generators in the WECC. Lastly, Battery cost reduc-

tions drive the most change. Both for generation mix and total cost of the system, battery cost reductions are capitalized on by the optimizer and result in the largest changes to the results for both datasets, but in particular the high electrification dataset when demand is higher.

Taken together, these results illustrate how the model responds to changes in technology costs and demand conditions, and they highlight the different ways each resource contributes to meeting a carbon free WECC. While the scenarios vary in total cost and in the deployment of geothermal, nuclear, and BESS, the broader layout of the system remains anchored by utility scale solar PV supported by wind, hydro, and storage technologies. The patterns observed across the 14 scenarios provide a clear foundation for interpreting how cost trajectories shape investment decisions and optimal system deployment. These findings encourage a deeper examination of what drives these results, the limitations of the modeling approach, and how the results answer the broader research question, all topics explored in the following Discussion section.

5 Discussion

5.1 Research Outcomes

The primary research question of this paper was focused on how changing costs of clean energy technologies will affect the optimal deployment of carbon free power systems in a capacity expansion model across the U.S. portion of the WECC. The results provide a clear, albeit

slightly surprising, answer. Generally reductions in costs for firm carbon free generators do not significantly change the optimal deployment of carbon free power systems in the WECC. Any major changes in this project were observed in how cost reductions for battery energy storage technologies affected optimal deployment. Battery cost reductions proved to be the most consequential driver of total system cost in this model, pushing reductions in cases where it was both the only technology with costs being reduced and in group scenarios. So much so that the single technology scenarios for non battery technologies are consistently more expensive than the all moderate scenarios, showcasing that even advanced reductions in geothermal and nuclear were less significant than the moderate reduction for BESS.

Why do major cost reductions in geothermal and nuclear fail to result in any major change to the interconnections capacity mix? This is due to the models setup and the inputs provided. EGS has a fixed O&M cost of \$135,226/ MWyear across all zones for both datasets. Nuclear also has a high fixed O&M cost of a little over \$100,000/ MWyear. Meanwhile the most dominant technology, solar utility PV, has costs of roughly \$12,000/ MWyear, a whole order of magnitude less than the other 2. Even with the most extreme cost reductions, EGS and nuclear still sit several tens of thousands of dollars per year higher than cheap VREs. This difference is not helped out by the fact that this models optimization function takes those metrics as key data points for the cost minimization function, meaning significant reductions are

likely closing in on a threshold to result in an explosion of deployment for firm generators, but still far enough outside it for it to not significantly impact results. Battery storage has a lower cost in the original datasets since this technology is significantly more developed and accepted, with higher amounts of baseline deployment also bumping the impact of any reductions. EGS reductions total at 0.105 \$B/year for the entire system in the most advanced case compared to the most expensive scenario. The advanced case for BESS reduces costs by 1.343 \$B/year, making the impact of battery cost reductions clear. While conservative cases for the 3 technologies are relatively similar, once the battery cost reduction goes above 50% a clear threshold is passed enabling greater deployment of BESS and a reduction of solar PV since energy generation from VRE can be better utilized by BESS.

Looking at the 3 ALL scenario trials it is worth noting that total cost reductions from conservative to advanced scenarios result in reductions of roughly 10% for both datasets. The high electrification scenarios are also fairly consistent with the increase of cost over the reference datasets, tracking between 4.9 - 6.2 \$B/year for these trials. This stability indicates that increased electrification demand is being met at a roughly constant margin cost regardless of the technology cost scenario.

The most striking finding comes from the result that BESS, rather than geothermal or nuclear, drives investment decisions in the model. In the reference dataset, installed battery capacity grows from 2.6 GW under EGS and nu-

clear single technology scenarios, where batteries are held at the dataset baseline, to 6.7 GW under EBSS conservative, 15.4 GW under BESS moderate, and 19.1 GW under BESS advanced. For the high electrification dataset, these figures scale dramatically; 23.3 GW, 26.6 GW, 41.6 GW, and 74.0 GW respectively. The 74 GW BESS capacity for the BESS advanced in the high electrification dataset represents a WECC in which storage has become the primary energy management mechanism at scale, absorbing the output of a large solar and wind fleet for daily and multi-day cycles. For both datasets annual battery discharge grows proportionally with the decrease in cost. The largest BESS discharge scenario actually comes from the Nuclear advanced trial for high electrification, where despite no present cost reductions for BESS, the system is its most active across all trials. The cause of this result is not immediately clear as the technology capacity mix actually doesn't change across any of the nuclear trials, the model just appears to dramatically increase BESS discharge in the nuclear advanced case resulting in further total system cost reductions over the prior cases.

Geothermal capacity results reveal a threshold behavior that is consistent with the structure of the dataset but should be carefully interpreted. In the reference and high electrification cases, geothermal installed capacity jumps from 0.64–0.66 GW under the conservative scenario to 1.65 GW under both Moderate and advanced, and then remains fixed at 1.65 GW across all higher cost reduction scenarios including all advanced and EGS advanced. The hold

out at 1.65 GW suggest that the model reaches a resource potential ceiling for geothermal resources, where further reasonable cost reductions beyond the moderate case cannot increase deployment because no additional resource potential is available in the dataset. This is an important result as the ceiling reflects the brown-field existing geothermal capacity in the WECC and not a representation of the full EGS resource potential that Ricks and Jenkins (2025) modeled. Their analysis projected up to 152 GW of deployable EGS nationally under favorable cost conditions, but that is not seen here. This result should not be taken as an argument that Ricks and Jenkins (2025) were in fact incorrect, with the reasoning for why firm resources may be undervalued in this model discussed in the Limitations subsection.

Despite these issues, geothermal generation varies notably even at fixed capacity. Annual geothermal generation rises from 5.3 TWh/yr under EGS conservative to 13.3 TWh/yr under EGS moderate in the reference case, despite the same 1.65 GW being installed. This demonstrates the model dispatching existing geothermal more intensively when its operational costs are reduced by the scenario multiplier. The slight decrease from 13.3 to 12.6 TWh/yr between EGS moderate and EGS advanced in the reference case is consistent with the model substituting cheap battery discharge for some geothermal output at the margin when both battery and geothermal costs are simultaneously reduced in the all scenarios.

The results for nuclear energy are also quite interesting. Nuclear installed capacity is iden-

tical across every scenario at 5.166 GW in both the reference and high electrification cases where the existing nuclear capacity (Palo Verde in WECC.NMAZ and the Columbia Generating Station in WECC.PNW) is held fixed for all trials. However, in the reference case, annual nuclear generation is just 0.06–0.12 TWh/yr across all scenarios. This implies a capacity factor of approximately 0.1–0.3 percent. Relative to the real world, where actual U.S. nuclear plants operate at capacity factors above 90 percent; a capacity factor of 0.2 percent means the model’s nuclear fleet is generating electricity for fewer than 18 hours per year, which is certainly not profitable for the nuclear plants. The economic interpretation is that in a system dominated by VRE sources with zero variable cost (i.e. solar, wind, geothermal, and hydro), the model has no incentive to dispatch nuclear at its variable cost of approximately \$2.3/MWh. This, in tandem with the minimum stable generation constraint (Min_power = 0.5), which would force significant output whenever nuclear is committed, results in low commitment by these generations, turning them into peaker generators rather than firm baseline power.

The final results come from the cross technology scenarios. The cross scenario nuclear and BESS Advanced achieves 12.655 \$B/yr in the reference case, which is 0.318 \$B/yr cheaper than BESS advanced alone (12.973 \$B/yr). This suggests that nuclear and battery cost reductions are mildly complementary where nuclear’s firm dispatchable generation reduces the peak shifting burden on batteries, increasing battery utilization and reducing the required bat-

tery fleet size. The EGS advanced with nuclear conservative scenario (13.634 \$B/yr) is 0.156 \$B/yr cheaper than EGS Advanced alone (13.790 \$B/yr), showcasing the limited role EGS plays due to its resource potential ceiling of 1.65 GW.

5.2 Limitations

This work is limited in a few key ways, primarily with regards to model fidelity, reliability metrics, simplified transmission, and cost only optimization. Examining first the lack of reliability metrics, this model does not include a valuation on firm capacity, or other values like Loss of load expectations (LOLE) or Effective Load Carrying Capability (ELCC). Those terms would help to more properly evaluate the adequacy of capacity build out to handle load demand on long time horizons. The lack of these may, and does, result in the overbuilding of VRE and BESS because the model does not see long duration reliability needs. Cost only optimizations assume any combination of resources that meet hourly demand in the representative dataset is reliable for functionality. This results in the undervaluation of firm resources that are already significantly more expensive, and causes a mismatch between optimal results and real world planning constraints. In practice, system planners assign higher capacity value to firm resources because they reduce risk of load shedding during extreme events. Without a mechanism to quantify this value, the model produces portfolios that appear cost optimal but could be operationally fragile. While the results of

this model show that major cost reductions in geothermal and nuclear fail to produce large deployment increases, it is unclear whether this reflects true economic inefficiency or the model's inability to recognize reliability metrics for firm resources.

The transmission network simplification to 6 zones for the data set with coarse interzonal transmission also fails to capture congestion within zones, land use constraints, and localized deliverability limits. This simplification likely plays a role in underestimating the cost of large VRE and BESS buildouts while simultaneously overestimating the ability of power to move freely across the WECC. Local bottlenecks could dramatically influence optimal portfolios.

Building on the previous point, the model allows for unlimited deployment of solar, wind, geothermal, and storage within some resource ceilings, but fails to examine land use conflicts. Especially for resources like wind and solar PV that require large tracts of land to serve an interconnection of this size, environmental and permitting constraints could limit the degree to which deployment occurs. Alongside this, geothermal energy also depends on subsurface conditions, limiting where it can be deployed and complicating the process for installation, especially in regions with earthquake fault lines like California. The expansion of transmission can also face constraints due to environmental constraints, with large interconnections often cutting through natural spaces that can result in long legal and planning battles to reach a resolution with local groups. While these constraints

can be difficult to quantify within a mathematical framework, they still represent important considerations and frames to place the results of this model in regard to.

Technological progress is also a complicated metric, and while this model simplifies progress to a simple multiplicative cost reducer, the real world is not so simple. Real world deployment can be shaped by tax credits, state subsidies, clean energy mandates, and market design allowing new and emerging technologies to break into the market even if costs still remain high. This in part limits the policy relevance of the results as there are more levers than just the raw economic cost that can be pulled to get increased deployment of these firm technologies. These results could suggest the importance of the utilization of these other levers to encourage deployment of geothermal and nuclear, but reductions of the prior limitations may be necessary to truly recommend this result.

5.3 Future Work

This model's objective function uses exclusively economical costs within the minimization function, and while this is an effective method for capacity expansion, in tandem with the research question it misses some real world influences that could have major impacts on the results. The lack of a steadier reliability metric to gauge capacity efficacy leaves some of the results of this project open ended. A top priority for future work on this model would be the addition of artificial weights. Specifically, weights that better demonstrate the value of firm generators

to grid planners that is not immediately clear in a cost minimization problem. Either by artificially adding costs to VRE to demonstrate that overbuilding with only VRE and BESS limits long term reliability in major grid events, or by artificially reducing costs for firm generators like geothermal and nuclear by quantifying that these generators do not rely on solar or wind availability to be able to serve load.

Additionally, these scenarios only examined how deployment would change if cost reductions came from technological progress. As mentioned before, in many Western states there exist mandates requiring high to complete carbon free grids in the coming decades. For states that truly value these goals, subsidies or tax breaks like those that boosted the solar industry in California, may come into play and reduce costs for geothermal and nuclear. Having scenarios that play into how subsidy or tax break cost reductions could further encourage the deployment of these technologies would add another cost reduction metric that could get these technologies over the hump to widespread commercial deployment.

A multi-year or multi-stage planning model could better capture feedback between deployment and cost reductions. Incorporating uncertainty through stochastic optimization would provide a more realistic view of investment risk. A more detailed representation of geothermal and nuclear technologies, including drilling success rates, modular construction timelines, or outage modeling, could refine their modeled roles. Also, since it is already included in the datasets and was simply not used, integrating

flexible load could broaden the set of pathways available to meet high electrification demand.

Lastly, the model has no way of measuring how land requirements and transmission could effect the results. While transmission costs are a part of the original model and the optimizer can expand transmission capacity if needed, there are only 6 zones with interzonal transmission not really considered. For example, the difficulties of interconnecting a massive solar field to a properly sized BESS deployment and then to electrical consumers could prove too difficult to build out the optimal capacity in this model, which could have a major impact on results. Increasing the amount of nodes and increasing transmission considerations would dramatically increase the complexity of this model, but would also stand to increase the applicability and specificity of the results in ways highly relevant to real life grid planning.

Overall, the results and accompanying analysis illustrate how the model responds to cost reductions across firm and flexible clean energy technologies, and how these responses shape the broader system design under both reference and high electrification futures. While the model's structure and assumptions influence the specific outcomes, the trends observed across the 14 scenarios provide a foundation for interpreting the role of technology learning, system flexibility, and resource constraints in a fully decarbonized WECC. These insights, along with the limitations and opportunities for future refinement outlined above, frame the broader implications of this work and motivate the following conclusion.

6 Conclusions

This study explored how cost uncertainties in enhanced geothermal systems, small modular nuclear reactors, and battery energy storage shape optimal investment decisions in a fully decarbonized WECC electricity system in 2045. Using a deterministic capacity expansion model and 14 technology cost scenarios under two demand datasets, the results demonstrate that system outcomes are sensitive to the relative cost trajectories of emerging clean energy technologies.

Across all scenarios, declining technology costs reduce total system expenditures, with the most pronounced effects observed in cases with lower BESS costs. Storage emerges as the most responsive and widely deployed technology, particularly under high electrification demand, where large increases in storage capacity and discharge are required to maintain reliability. Geothermal deployment occurs only when cost reductions exceed roughly 50%, while nuclear capacity remains largely unchanged across all scenarios, indicating limited sensitivity to cost variation within the modeled range.

Despite these differences, the broader system configuration remains dominated by solar PV, supported by wind and hydro resources, with storage providing the flexibility needed to balance variable renewable output. High electrification scenarios consistently require greater total capacity and significantly more storage, showcasing the increased challenge of meeting higher loads with carbon free resources.

Overall, the results highlight the importance

of technology learning rates in shaping future clean energy portfolios and underscore the value of continued cost reductions in storage and firm carbon free technologies. Continued work could improve applicability of these results with the inclusions of reliability metrics, other cost reducing measures like subsidies, and increasing the granularity of the modeled system to better capture transmission and land usage in the WECC. These findings provide insight into how investment priorities and policy support may influence the evolution of a reliable, affordable, and fully decarbonized WECC grid.

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Appendix: Nomenclature

Variables

- x_g^{new} New power capacity built (MW)
- x_g^{cap} Total installed power capacity (existing – retired + new) (MW)
- e_g^{new} New storage energy capacity built (MWh)
- e_g^{cap} Total storage energy capacity (MWh)
- T_l^{cap} Total transmission capacity on line l (MW)
- T_l^{new} New transmission capacity added to line l (MW)
- $P_{g,t}$ Power dispatch of generator g at hour t (MW)
- $u_{s,z,t}$ Non-served energy in segment s , zone z , hour t (MW)

Objective and Cost Components

Sets

Z Total annualized system cost (\$/yr)

$g \in \mathcal{G}$ Set of generators

Z_{fixed} Total fixed investment and fixed O&M cost (\$/yr)

$e \in \mathcal{E}$ Set of storage resources

$l \in \mathcal{L}$ Set of transmission lines

Z_{var} Total variable operating cost (\$/yr)

$z \in \mathcal{Z}$ Set of zones

Z_{NSE} Total non-served energy cost (\$/yr)

$s \in \mathcal{S}$ Set of demand/NSE segments

$Z_{\text{transmission}}$ Total transmission cost (\$/yr)

$t \in \mathcal{T}$ Set of modeled hours

Parameters

C_g^{inv} Annualized investment cost (\$/MWyr)

C_g^{fom} Fixed operations & maintenance cost (\$/MWyr)

$C_g^{\text{inv,e}}$ Annualized storage energy investment cost (\$/MWyr)

$C_g^{\text{fom,e}}$ Fixed O&M on storage energy capacity (\$/MWyr)

C_g^{var} Variable O&M + fuel cost per MWh generated (\$/MWh)

C_l^{fix} Fixed cost of existing transmission capacity (\$/MWyr)

C_l^{inv} Investment cost of new transmission capacity (\$/MWyr)

w_t Sample weight (annual hours represented by modeled hour t) (h/yr)

V_s Value of lost load for NSE segment s (\$/MW)